Homework 08

Question 1: Boundary and margin

```
d20 <- read.csv("~/Desktop/stat432-fall2026/week-08/data/margin-geometry-20.csv")
beta <- c(0.8, 0.6)
tol  <- 1e-8

d20$f        <- as.numeric(as.matrix(d20[, c("x1", "x2")]) %*% beta)
d20$distance <- abs(d20$f) / sqrt(sum(beta^2))
d20$margin   <- d20$y * d20$f
d20$position <- ifelse(abs(d20$margin - 1) <= tol, "on",
                ifelse(d20$margin > 1, "beyond", "inside"))
d20$position <- factor(d20$position, levels = c("inside", "on", "beyond"))
d20$pred     <- ifelse(d20$f > 0, 1, -1)
d20$misclassified <- d20$pred != d20$y
```

(a) Distances, functional margins, and margin position

An observation is **beyond** the margin when $yf(x) > 1$, **on** the margin when $yf(x) = 1$, and **inside** the margin when $yf(x) < 1$. Because $\|(0.8, 0.6)\|_2 = 1$, the distance to the boundary is $|f(x)|$.

```
tab <- d20[, c("observation", "x1", "x2", "y", "f", "distance", "margin", "position")]
names(tab)[7] <- "y*f(x)"
kable(tab, digits = 3, row.names = FALSE)
```

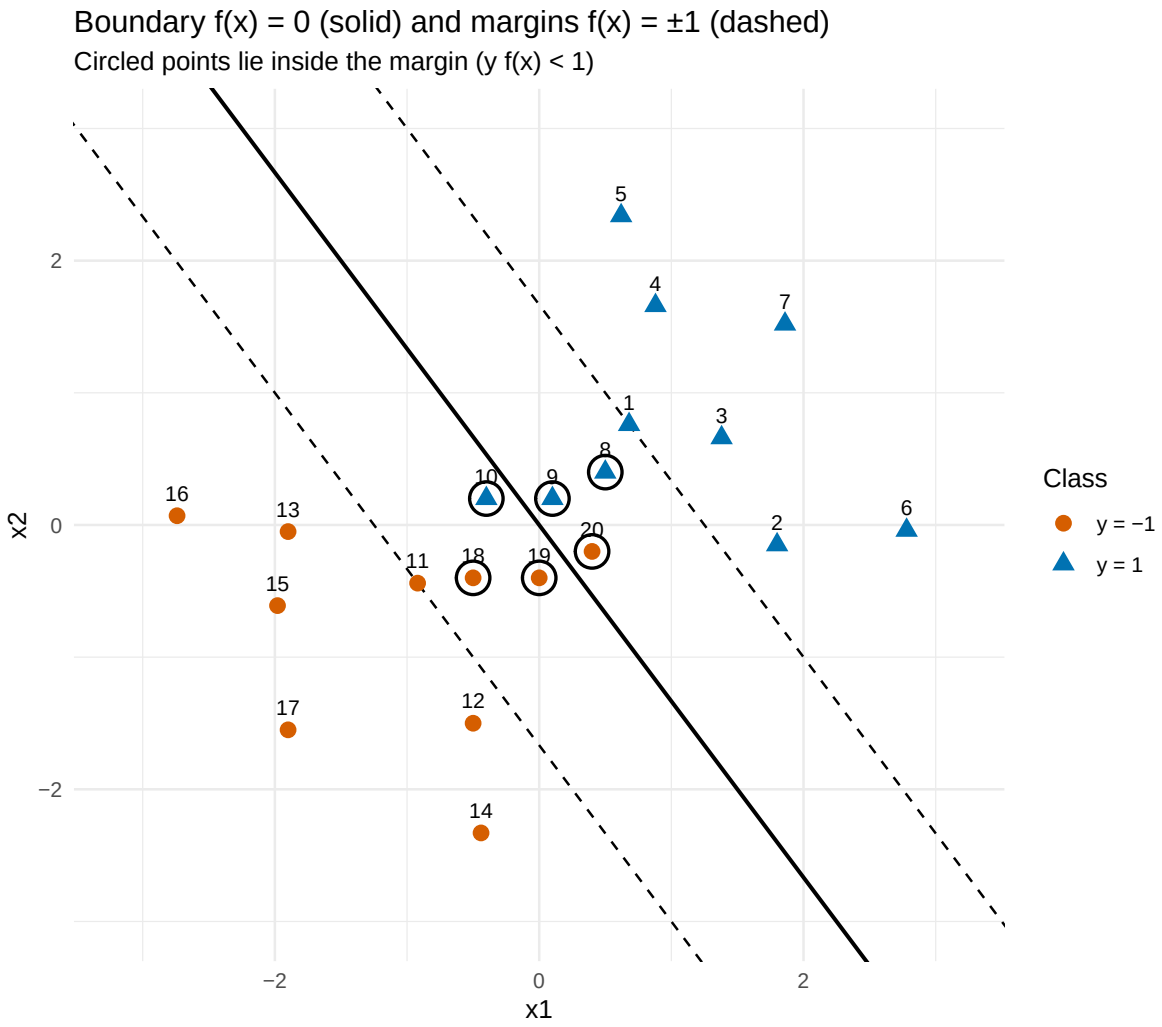
observation	x1	x2	y	f	distance	y*f(x)	position
1	0.68	0.76	1	1.00	1.00	1.00	on
2	1.80	-0.15	1	1.35	1.35	1.35	beyond

observation	x1	x2	y	f	distance	y*f(x)	position
3	1.38	0.66	1	1.50	1.50	1.50	beyond
4	0.88	1.66	1	1.70	1.70	1.70	beyond
5	0.62	2.34	1	1.90	1.90	1.90	beyond
6	2.78	-0.04	1	2.20	2.20	2.20	beyond
7	1.86	1.52	1	2.40	2.40	2.40	beyond
8	0.50	0.40	1	0.64	0.64	0.64	inside
9	0.10	0.20	1	0.20	0.20	0.20	inside
10	-0.40	0.20	1	-0.20	0.20	-0.20	inside
11	-0.92	-0.44	-1	-1.00	1.00	1.00	on
12	-0.50	-1.50	-1	-1.30	1.30	1.30	beyond
13	-1.90	-0.05	-1	-1.55	1.55	1.55	beyond
14	-0.44	-2.33	-1	-1.75	1.75	1.75	beyond
15	-1.98	-0.61	-1	-1.95	1.95	1.95	beyond
16	-2.74	0.07	-1	-2.15	2.15	2.15	beyond
17	-1.90	-1.55	-1	-2.45	2.45	2.45	beyond
18	-0.50	-0.40	-1	-0.64	0.64	0.64	inside
19	0.00	-0.40	-1	-0.24	0.24	0.24	inside
20	0.40	-0.20	-1	0.20	0.20	-0.20	inside

(b) Plot

```
d20$class  <- factor(d20$y, levels = c(-1, 1), labels = c("y = -1", "y = 1"))
d20$inside <- d20$position == "inside"

ggplot(d20, aes(x1, x2)) +
  geom_abline(intercept =  0,       slope = -0.8/0.6, linewidth = 0.8) +
  geom_abline(intercept =  1/0.6,   slope = -0.8/0.6, linetype = "dashed") +
  geom_abline(intercept = -1/0.6,   slope = -0.8/0.6, linetype = "dashed") +
  geom_point(aes(color = class, shape = class), size = 3) +
  geom_point(data = subset(d20, inside), shape = 1, size = 6, stroke = 1) +
  geom_text(aes(label = observation), nudge_y = 0.17, size = 3) +
  scale_color_manual(values = c("y = -1" = "#D55E00", "y = 1" = "#0072B2")) +
  coord_fixed(ratio = 1, xlim = c(-3.2, 3.2), ylim = c(-3, 3)) +
  labs(title = "Boundary f(x) = 0 (solid) and margins f(x) = ±1 (dashed)",
       subtitle = "Circled points lie inside the margin (y f(x) < 1)",
       color = "Class", shape = "Class") +
  theme_minimal()
```



(c) Counts and misclassifications

```
kable(as.data.frame(table(Position = d20$position)), col.names = c("Position", "Count"))
```

Position	Count
inside	6
on	2
beyond	12

```r
inside_ids <- d20$observation[d20$position == "inside"]
mis_ids    <- d20$observation[d20$misclassified]
inside_ids
```

```
[1]  8  9 10 18 19 20
```

```r
mis_ids
```

```
[1] 10 20
```

Among the 20 observations, 6 are inside the margin, 2 are on the margin, and 12 are beyond it. The rule $\hat{y} = \text{sign}(f(x))$ misclassifies observations 10 and 20, which have $yf(x) < 0$ (they sit on the wrong side of the boundary). Observations 8, 9, 18, 19 are correctly classified but lie inside the margin ($0 < yf(x) < 1$).

Question 2: Margin in four dimensions

```r
d4   <- read.csv("~/Desktop/stat432-fall2026/week-08/data/margin-geometry-4d.csv")
X    <- as.matrix(d4[, c("x1", "x2", "x3", "x4")])
b0   <- -1
b    <- c(1, 2, -2, 4)
nb   <- sqrt(sum(b^2))

d4$f     <- as.numeric(b0 + X %*% b)
d4$proj  <- as.numeric(X %*% b) / nb
d4$dist  <- d4$f / nb
d4$fm    <- d4$y * d4$f
d4$class <- factor(d4$y, levels = c(-1, 1), labels = c("y = -1", "y = 1"))
boundary <- -b0 / nb
c(norm_beta = nb, boundary = boundary, misclassified = sum(d4$fm <= 0))
```

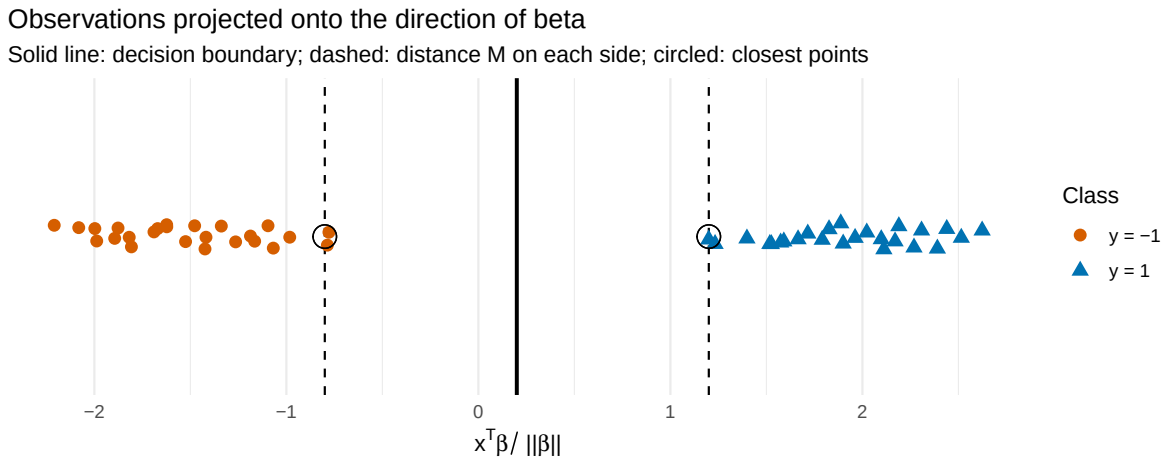
```
    norm_beta      boundary misclassified
          5.0           0.2           0.0
```

(a) Projection plot and geometric margin

The boundary $f(\mathbf{x}) = 0$ corresponds to the projection $\mathbf{x}^\top \beta / \|\beta\| = -\beta_0 / \|\beta\| = 1/5 = 0.2$.

```
M <- min(d4$y * d4$dist)
closest <- d4[abs(d4$y * d4$dist - M) < 1e-8, ]

ggplot(d4, aes(proj, 0, color = class, shape = class)) +
  geom_vline(xintercept = boundary, linewidth = 0.9) +
  geom_vline(xintercept = boundary + c(-M, M), linetype = "dashed") +
  geom_point(size = 2.6, position = position_jitter(height = 0.05, seed = 1)) +
  geom_point(data = closest, size = 5, shape = 1, color = "black") +
  scale_color_manual(values = c("y = -1" = "#D55E00", "y = 1" = "#0072B2")) +
  scale_y_continuous(limits = c(-0.5, 0.5), breaks = NULL) +
  labs(x = expression(x^T * beta / ~ "||" * beta * "||"), y = NULL,
       title = "Observations projected onto the direction of beta",
       subtitle = "Solid line: decision boundary; dashed: distance M on each side; circled: c
       color = "Class", shape = "Class") +
  theme_minimal()
```




```
kable(head(d4[order(d4$fm), c("observation", "y", "f", "dist", "fm")], 6), digits = 3,
      row.names = FALSE)
```

observation	y	f	dist	fm
4	1	5	1.0	5
15	-1	-5	-1.0	5
33	1	5	1.0	5
36	-1	-5	-1.0	5
17	-1	-6	-1.2	6
31	1	6	1.2	6

```
M
```

```
[1] 1
```

The classes are perfectly separated (no observation has $yf(\mathbf{x}) \leq 0$). The closest observations (4, 15, 33, 36) all have $|f(\mathbf{x})| = 5$, so their distance to the boundary is $5/\|\beta\| = 5/5$. Hence the **geometric margin is** $M = 1$.

(b) SVM normalization

The smallest functional margin is $\min_i y_i f(\mathbf{x}_i) = 5$. Dividing β_0 and β by 5 leaves the boundary unchanged and sets the smallest functional margin to 1:

```
min_fm <- min(d4$fm)
beta_svm  <- b / min_fm
beta0_svm <- b0 / min_fm
beta_svm; beta0_svm
```

```
[1]  0.2  0.4 -0.4  0.8
```

```
[1] -0.2
```

```
min(d4$y * (beta0_svm + X %*% beta_svm)); 1 / sqrt(sum(beta_svm^2))
```

```
[1] 1
```

```
[1] 1
```

$$\beta_{\text{SVM}} = (0.2,\ 0.4,\ -0.4,\ 0.8)^{\mathsf{T}}, \qquad \beta_{0,\text{SVM}} = -0.2.$$

Then $\|\beta_{\text{SVM}}\| = 1$, consistent with $M = 1/\|\beta_{\text{SVM}}\| = 1$.

Question 3: Peer review of a Homework 7 skill

Your review:

Overall assessment. The skill is a well-built, leakage-safe pipeline. It reserves the test set first, reuses fixed CV folds, keeps scaling and PCA inside Pipelines, and uses the test set once. It is also easy to follow, and Rahaf's report matches what the skill specifies. When I ran it with her seed (4323), I reproduced every number in her Q4 table. The weaknesses are in the underspecified rules and in how little the single-split design can support.

1. The seed rule contradicts itself.

Observed: The skill says to ask before starting if anything is missing, but it also lists the seed as optional. I choose 432. Why it matters: The seed changed the result. With 432 the selected model was PCA + LDA with d = 4 and a test error of 0.252. With 4323 it was d = 5 and 0.338. What to change: State the default explicitly, for example "if no seed is given, use a fixed default and say so in the report." Alternatively, make the seed required.

2. "Simpler candidate" is not defined.

Observed: The skill says "LDA before QDA, fewer components" but gives no total order. Is plain LDA simpler than PCA + LDA with d = 4? Is a blend simpler than PCA + QDA? I invented an ordering, and under seed 432 it decided the outcome (d = 4 was chosen over a lower-error d = 7). Why it matters: Two people following the skill could select different models from the same CV table, which undermines reproducibility. What to change: Give an explicit ranking, for example by number of estimated parameters, and use it for ties.

3. One split gives an unstable selection and test error.

Observed: One 5-fold split and one 30% test set (139 cases, 48 with CHD) gave a selected-model test error of 0.252 under seed 432 and 0.338 under 4323. That gap is more than two binomial SEs of about 0.04. The best PCA dimension also moved among d = 4, 5 and 7. Her ranking against plain LDA and QDA flipped too: the selected model is worse than both under 4323 and better than both under 432. Why it matters: The report's headline number is mostly split luck, and the bullet on how much the extensions helped is not stable across seeds. The SD / sqrt(5) SE also ignores correlation between folds, so it is optimistic. What to change: Use repeated stratified CV for selection. Require a sensitivity line in the report (for example, selected model and test error across 3 to 5 seeds), or state clearly that the result is conditional on one seed.

About the run itself. With seed 432: n = 462, p = 9, 323 train and 139 test; selected PCA + LDA d = 4, CV 0.266 (SE 0.022), test error 0.252 (35 of 139); plain LDA and QDA test errors 0.266 and 0.281; no failed fits. With seed 4323: d = 5, CV 0.235 (SE 0.012), test error 0.338; plain LDA and QDA 0.324 and 0.317, matching her report.

GitHub Issue link: https://github.com/rahafala/stat432-fall2026/issues/4